Course	
Term	
Week	
Date	
Chapter. Topic	PCEP Overview

Python Certified Entry Level Programmer (PCEP) Overview

Siva R Jasthi

www.learnandhelp.com

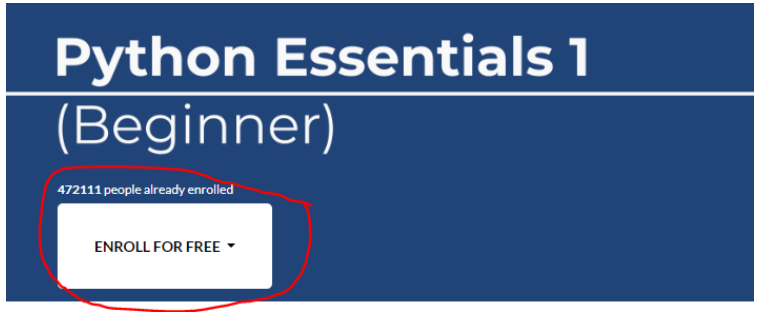
Registration Logistics

[1] Register at https://edube.org/
Use the student's (certification candidate) name and email while registering.

[2] Select the PE1 course
https://edube.org

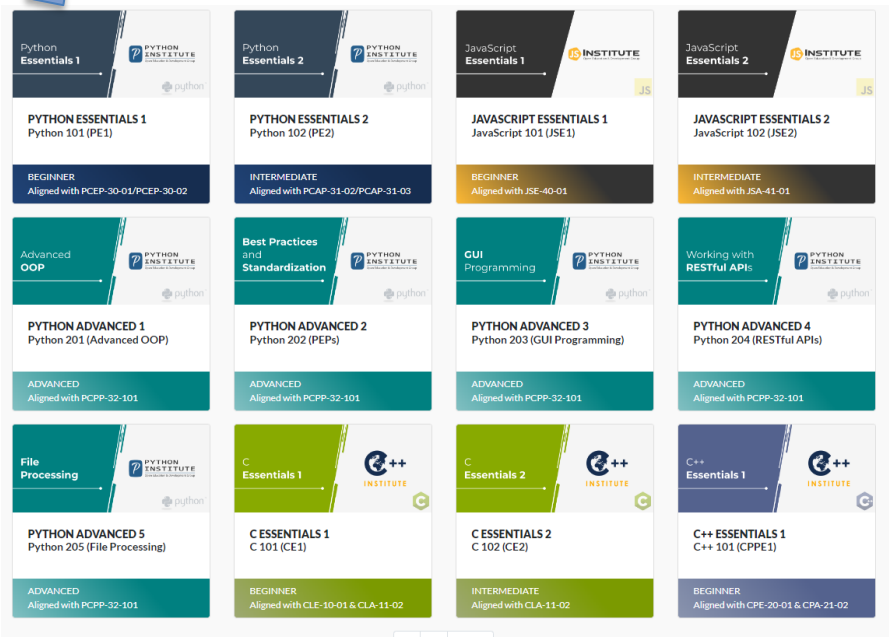
[3] Enroll in the PE1 course

https://edube.org/study/pe1



Python 101 (PE1) @ edube.org

https://edube.org/study/pe1



Certifications

PCEP – Certified Entry-Level Python Programmer

PCAP – Certified Associate in Python Programming

PCPP1 – Certified Professional in Python Programming 1

Certification

PCEP – Certified Entry-Level Python Programmer certification is a professional credential that measures your ability to accomplish coding tasks related to the essentials of programming in the Python language. A test candidate should demonstrate sufficient knowledge of the universal concepts of computer programming, the syntax and semantics of the Python language as well as the skills in resolving typical implementation challenges with the help of the Python Standard Library.

Python Essentials 1

Python Essentials 1 (Aligned with PCEP-30-02)

Welcome to PE1	PE1: Module 1	PE1: Module 2	PE1: Module 3	PE1: Module 4	PE1 Summary Test	PCEP-30-01 Exam
About the course ✔	Study Resources ✔	Study Resources ✔	Study Resources ✔	Study Resources ✔	End-of-course Information ✔	
	Quiz: 100%	Quiz: 100%	Quiz: 100%	Quiz: 75%	Summary Test: 83%	
	Test: 100%	Test: 95%	Test: 90%	Test: 82%		

Last activity - 2023-04-02 04:24:43

Resume Course

FREE

BUY THE VOUCHER

Course and Exam

1. Course and practice tests are free to take.

2. But “Exam” costs money

Strategy:

[1] Complete the course first.
[2] Take practice quizzes and tests along the way
[3] Once you complete all the modules (there are four), take the comprehensive practice test.
[4] When you are ready, then buy the “Exam Voucher” and take the Exam right away.

Exam: Online exam and you can take it anytime.

Channel: Edube Interactive

Sponsor: Python Institute

Mode: Online self-study course

Level: Beginner

Cost: Free

Study Time: 42 hours (Recommended: 7h/week)

Language: English, Spanish

Associated Certification: PCEP – Certified Entry-Level Python Programmer (Exam PCEP-30-01 & PCEP-30-02)

Buy Exam Voucher

Course is free

Course Syllabus

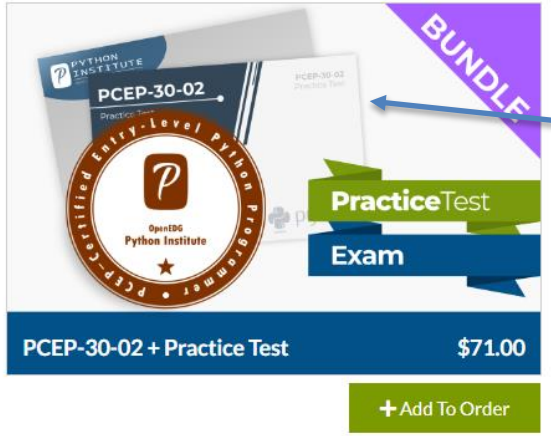
1. Introduction to Python and computer programming
2. Data types, variables, basic input-output operations, basic operators
3. Boolean values, conditional execution, loops, lists, logical and bitwise operations
4. Functions, tuples, dictionaries, and data processing

We already covered much of this course (and many additional topics) in our Python 101 course for the last 25 weeks

However, you need to do three things here:

- Review the material as a refresher
- Take the practice quizzes in each module
- Take the comprehensive tests at the end of the course

Exam Fee



I recommend the second BUNDLE $71.00 (Exam + Practice Test) if you score above 80% in your practice tests.

If you score less than 80% on the practice tests, do not buy the exam until you get better.

Exam Format

PCEP™: Exam Information

SPECIFICATIONS ITEM	DESCRIPTION
Exam name	PCEP™ – Certified Entry-Level Python Programmer
Exam Code & Current Exam Versions	PCEP-30-02 (Status: Active) PCEP-30-01 (Status: Retiring – December 31, 2022)
Prerequisites	None
Validity	Lifetime
Exam Duration	PCEP-30-02 – Exam: 40 minutes, NDA/Tutorial: 5 minutes PCEP-30-01 – Exam: 45 minutes, NDA/Tutorial: 5 minutes
Number of Questions	30
Format	Single- and multiple-select questions, drag & drop, gap fill, sort, code fill, code insertion | Python 3.x
Passing Score	70%
Languages	English, Spanish
Cost	USD 59 (Exam: Single-Shot) USD 76.70 (Exam: with one retake) USD 71.00 (Exam: Single-Shot + Practice Test) USD 29 (Practice Test)

PCEP Certification: Study Plan

[1] Focus on one module per day (4 * 1 = 4 days = 3 hours each)

- Review the material (Python 101)
- Take WWPP quiz (question bank for Python 101)
- Take the practice quiz
- Take the Practice Test

Keep taking these quizzes and tests till you score 100% in each of these.

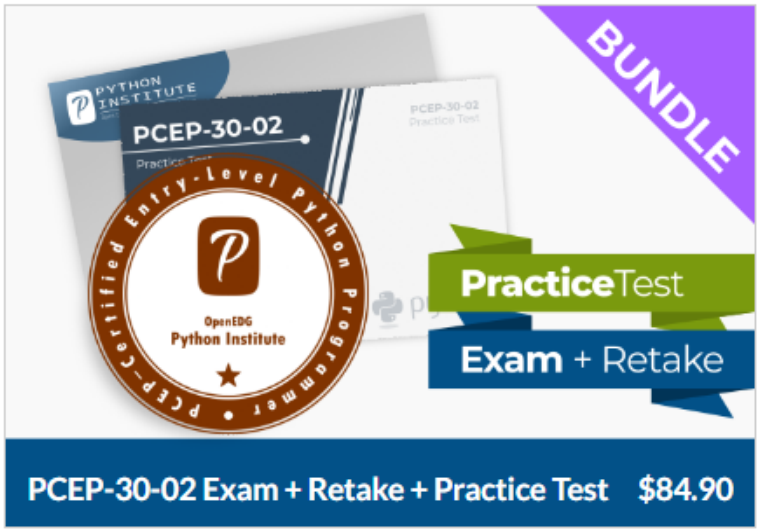
[2] Take the final Summary test (1 day) ~= 1 hour)

- If you score above 90%, you are ready for the PCEP.

[3] Buy this package once you complete steps 1 and 2
Exam + Retake + Practice Test ($84)

[4] Take the Practice Exam
You can retake this till you score 90% or above

[5] and then take the PCEP exam
I recommend taking the practice exam and the real exam on the same day



Python Essentials 1 = 4 Modules

The course is divided into 4 modules:

1. **Module 1**
   Introduction to Python and computer programming;
2. **Module 2**
   Data types, variables, basic input-output operations, and basic operators;
3. **Module 3**
   Boolean values, conditional execution, loops, lists and list processing, logical and bitwise operations;
4. **Module 4**
   Functions, tuples, dictionaries, exceptions, and data processing.

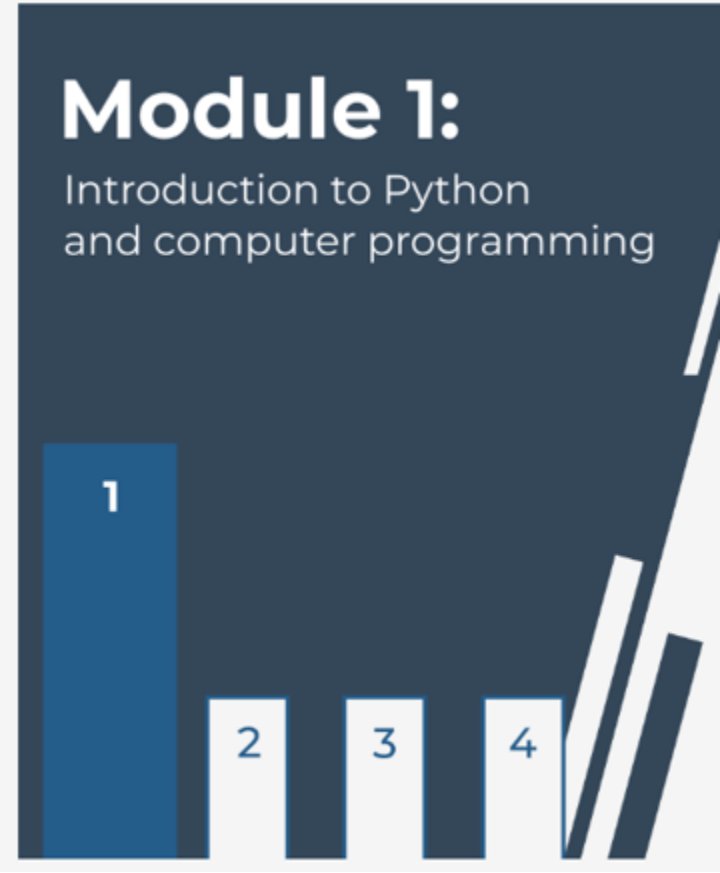
Module 1

Python Essentials 1:
Module 1

Introduction to Python and computer programming

In this module, you will learn about:

- the fundamentals of computer programming, i.e., how the computer works, how the program is executed, how the programming language is defined and constructed;
- the difference between compilation and interpretation;
- what Python is, how it is positioned among other programming languages, and what distinguishes the different versions of Python.



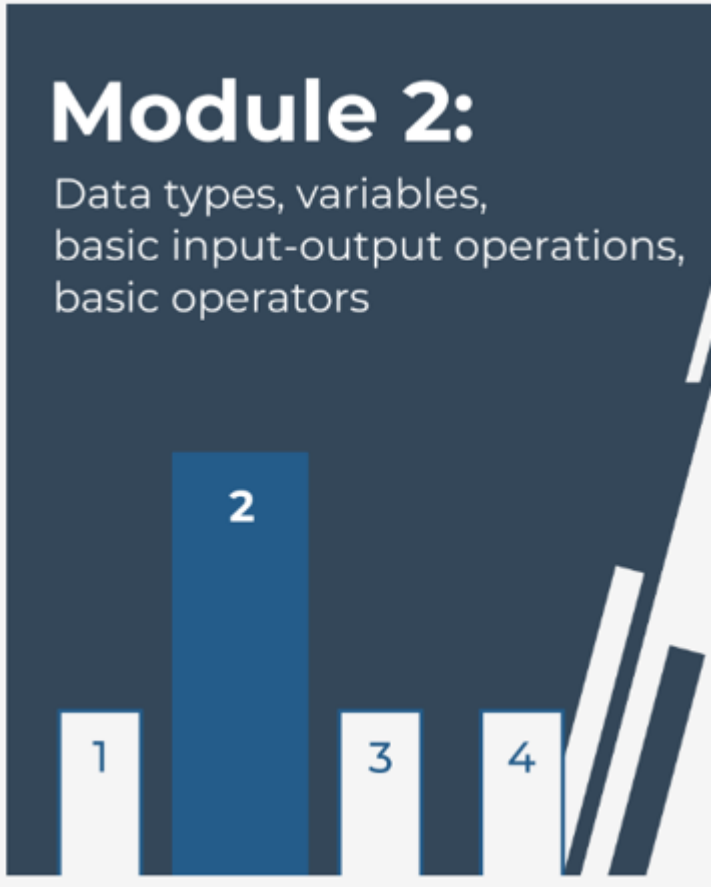
Module 2

Python Essentials 1:
Module 2

Data types, variables, basic input-output operations, basic operators

In this module, you will learn:

- how to write and run simple Python programs;
- what Python literals, operators, and expressions are;
- what variables are and what are the rules that govern them;
- how to perform basic input and output operations.



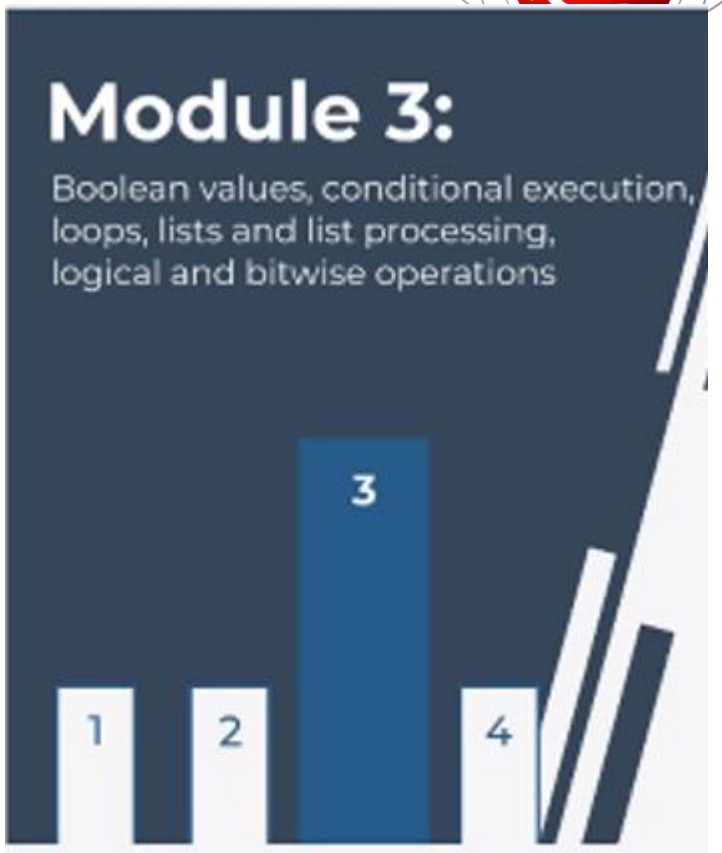
Module 3

Python Essentials 1: Module 3

Boolean Values, Conditional Execution, Loops, Lists and List Processing, Logical and Bitwise Operations

In this module, you will cover the following topics:

- the Boolean data type;
- relational operators;
- making decisions in Python (if, if-else, if-elif,else)
- how to repeat code execution using loops (while, for)
- how to perform logic and bitwise operations in Python;
- lists in Python (constructing, indexing, and slicing; content manipulation)
- how to sort a list using bubble-sort algorithms;
- multidimensional lists and their applications.



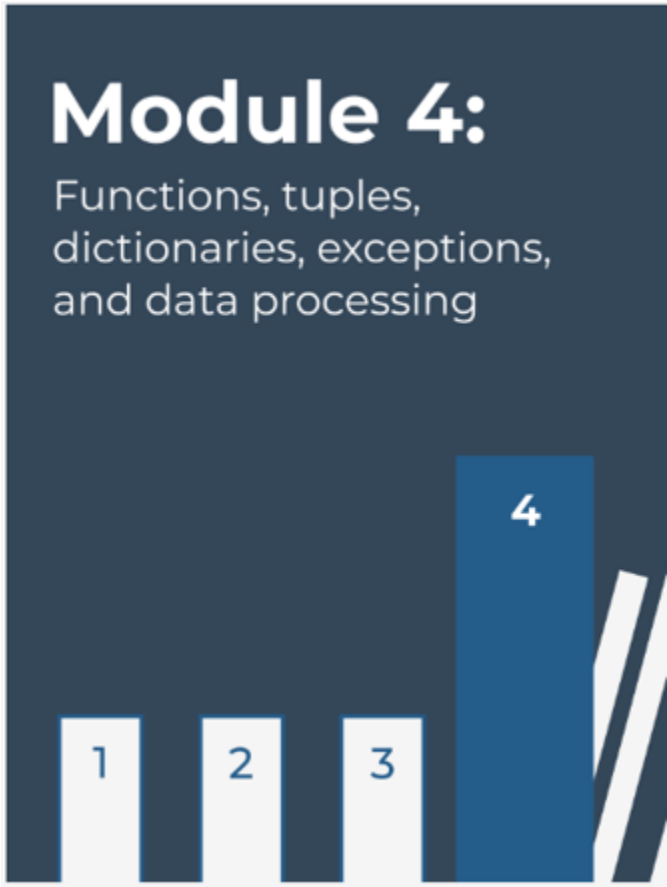
Module 4

Python Essentials 1:
Module 4

Functions, Tuples, Dictionaries, Exceptions, and Data Processing

In this module, you will cover the following topics:

- code structuring and the concept of function;
- function invocation and returning a result from a function;
- name scopes and variable shadowing;
- tuples and their purpose, constructing and using tuples;
- dictionaries and their purpose, constructing and using dictionaries;
- exceptions – the *try* statement and the *except* clause, Python built-in exceptions, code testing and debugging.



PCEP: Quizzes and Tests (Module 1)

PE1 -- Module 1 Quiz

https://edube.org/quiz/pe-1/pe1-module-1-quiz-1

Time limit: 15 minutes
Number of questions: 10
Points to score: 10
Passing score: 70%

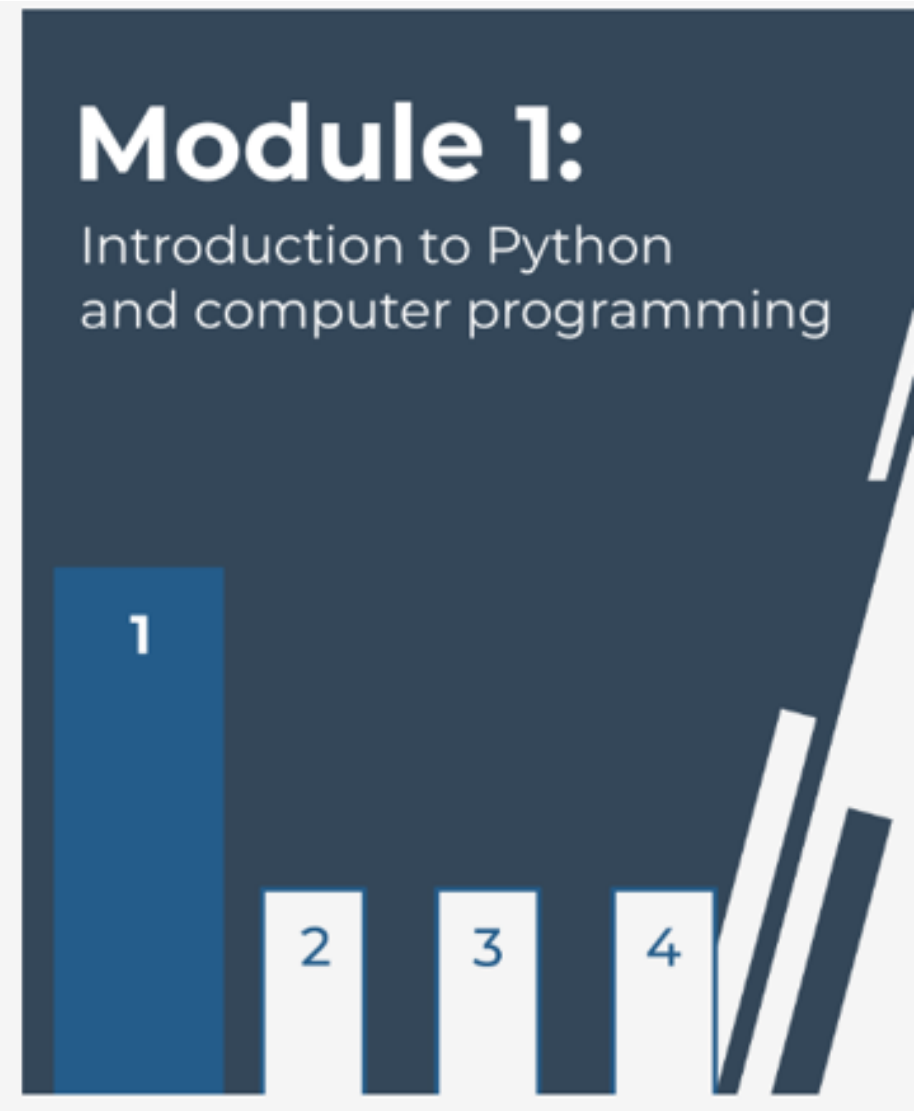
Start

PE1 -- Module 1 Test

https://edube.org/quiz/pe-1/pe1-module-1-test-1

Time limit: 20 minutes
Number of questions: 10
Points to score: 10
Passing score: 70%

Start



PCEP: Quizzes and Tests

PE1 -- Module 2 Quiz

https://edube.org/quiz/pe-1/pe1-module-2-quiz-1

Time limit: 15 minutes
Number of questions: 10
Points to score: 10
Passing score: 70%

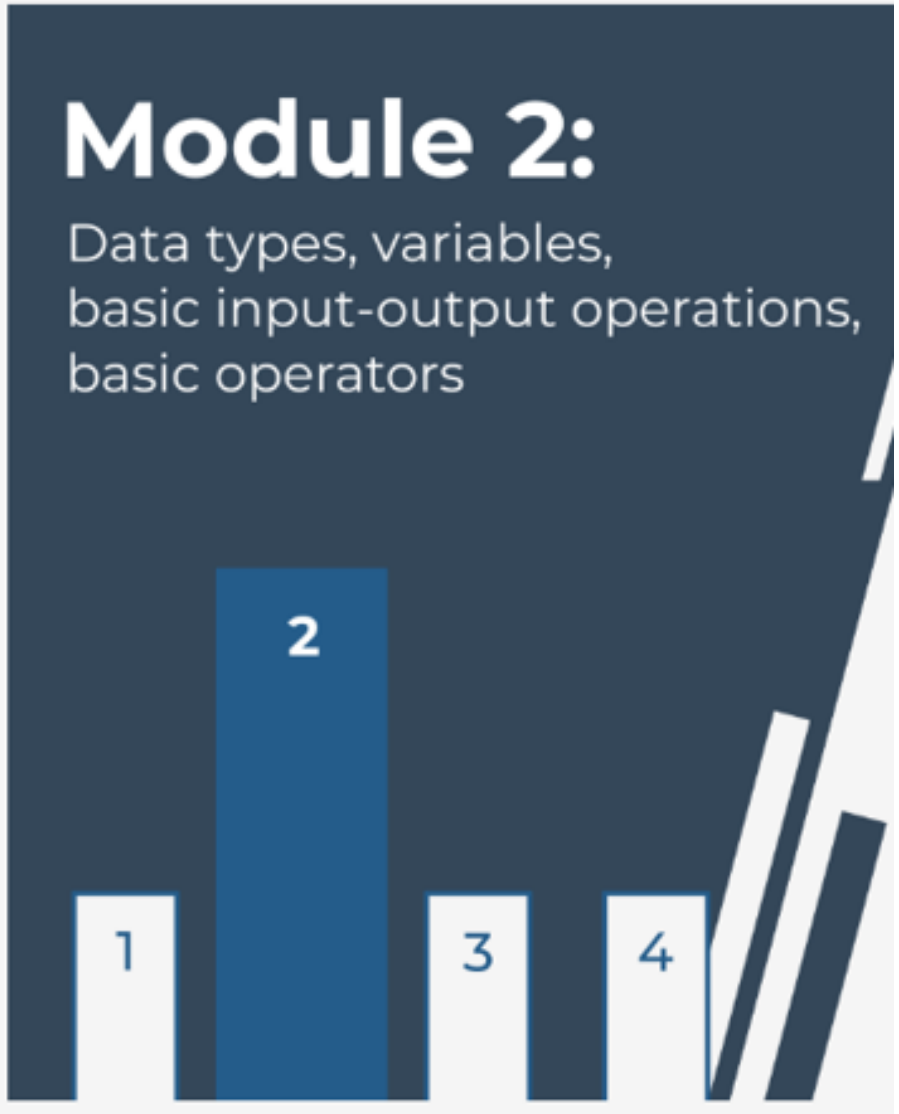
Start

PE1 -- Module 2 Test

https://edube.org/quiz/pe-1/pe1-module-2-test-1

Time limit: 30 minutes
Number of questions: 20
Points to score: 20
Passing score: 70%

Start



PCEP: Quizzes and Tests

PE1 -- Module 3 Quiz

https://edube.org/quiz/pe-1/pe1-module-3-quiz-1

Time limit: 15 minutes
Number of questions: 10
Points to score: 10
Passing score: 70%

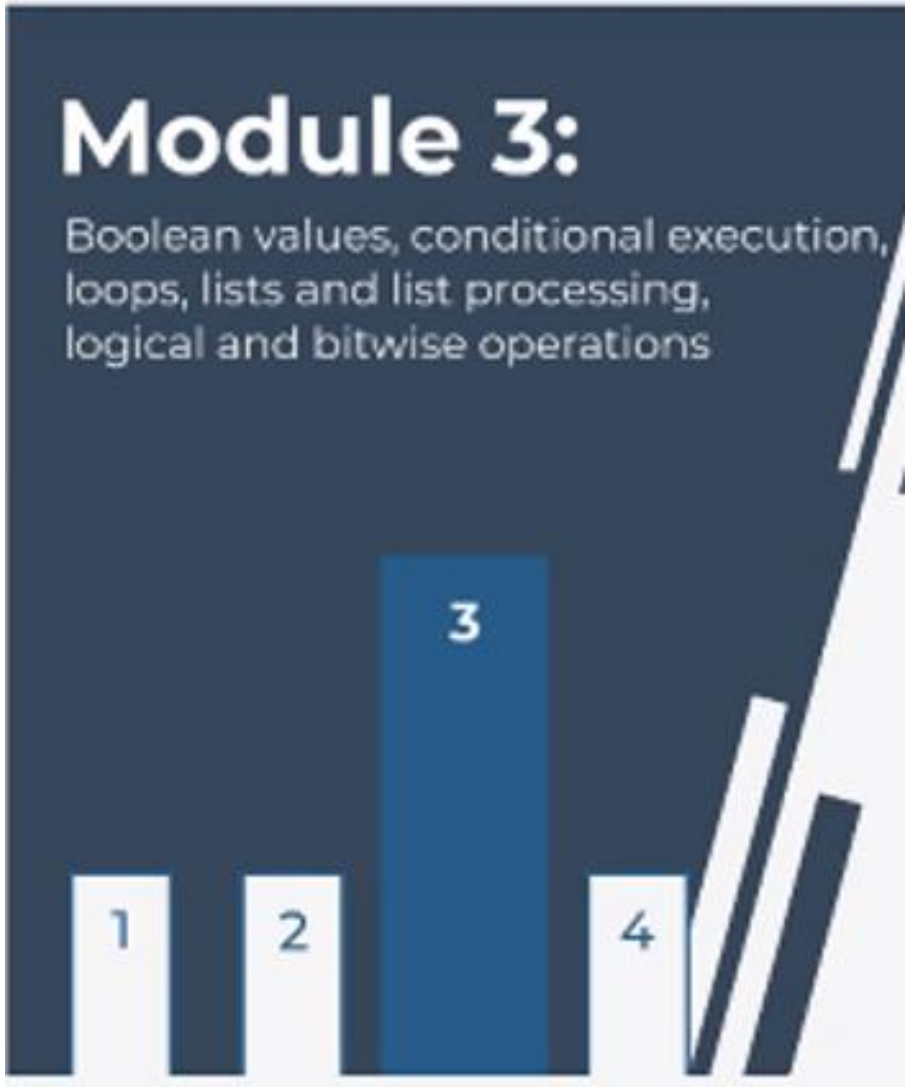
Start

PE1 -- Module 3 Test

https://edube.org/quiz/pe-1/pe1-module-3-test-1

Time limit: 30 minutes
Number of questions: 20
Points to score: 20
Passing score: 70%

Start



PCEP: Quizzes and Tests

PE1 -- Module 4 Quiz

https://edube.org/quiz/pe-1/pe1-module-4-quiz-1

Time limit: 15 minutes
Number of questions: 12
Points to score: 12
Passing score: 70%

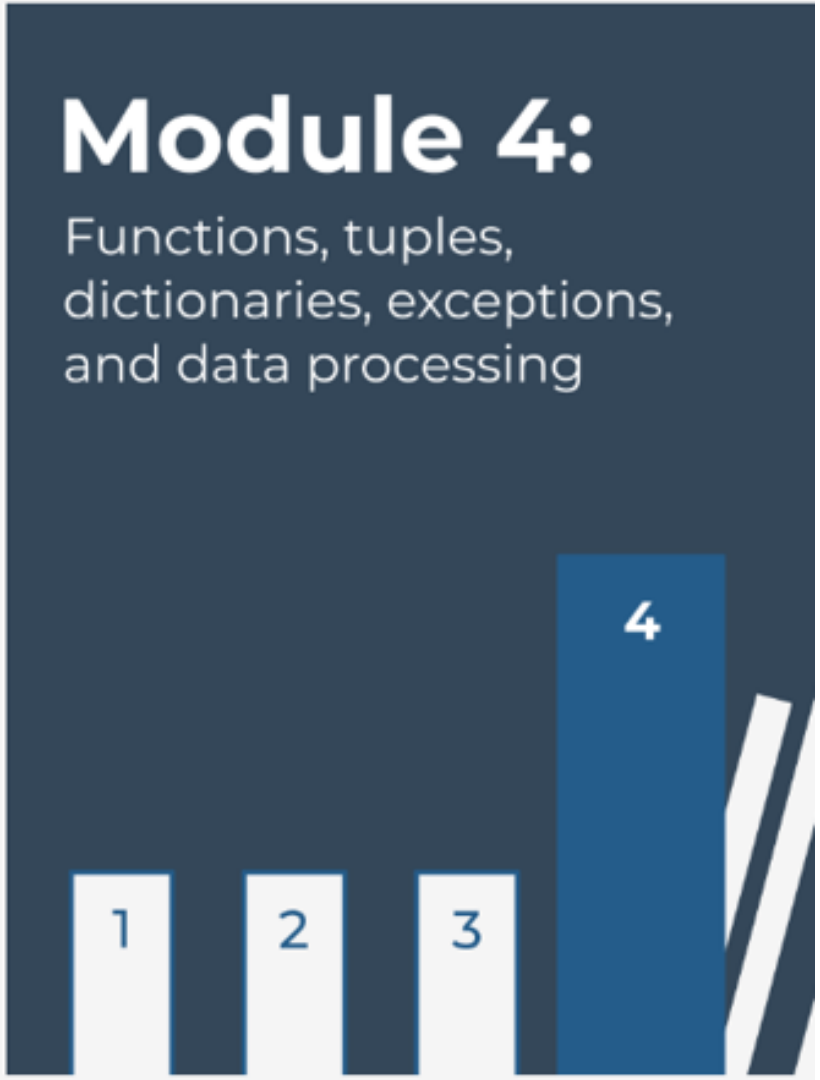
Start

PE1 -- Module 4 Test

https://edube.org/quiz/pe-1/pe1-module-4-test-1

Time limit: 30 minutes
Number of questions: 22
Points to score: 22
Passing score: 70%

Start



PCEP: Quizzes and Tests

PE1 -- Summary Test

https://edube.org/quiz/pe-1/pe1-summary-test-1

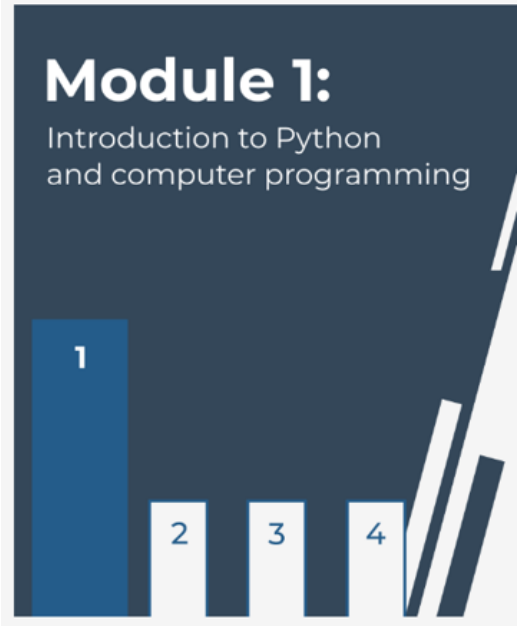
Time limit: 45 minutes

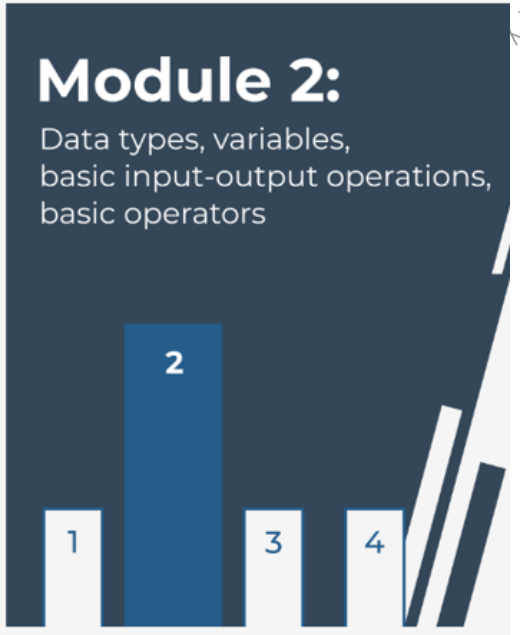
Number of questions: 35

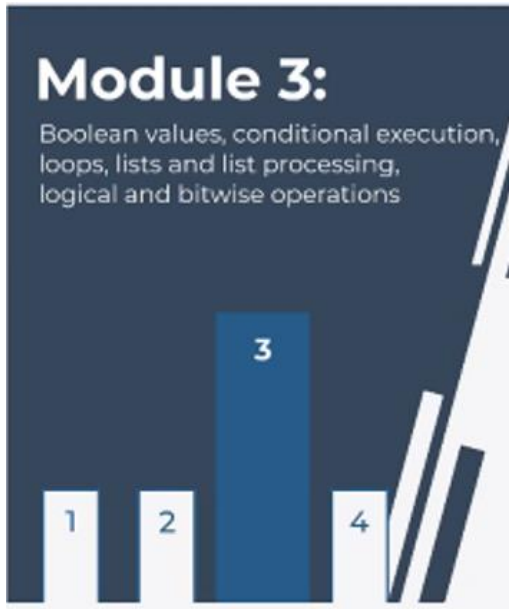
Points to score: 35

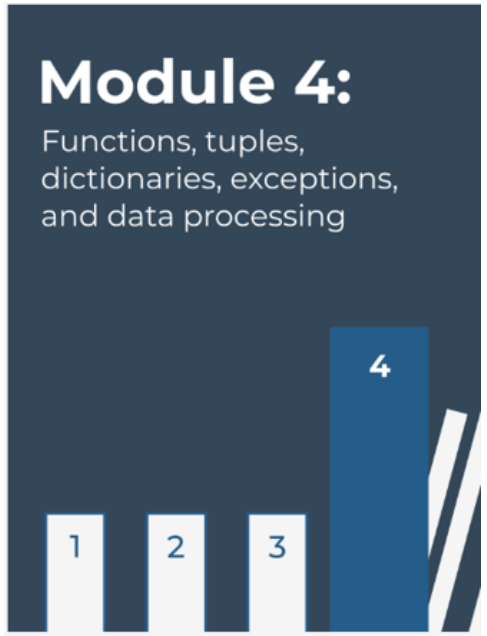
Passing score: 70%

Start



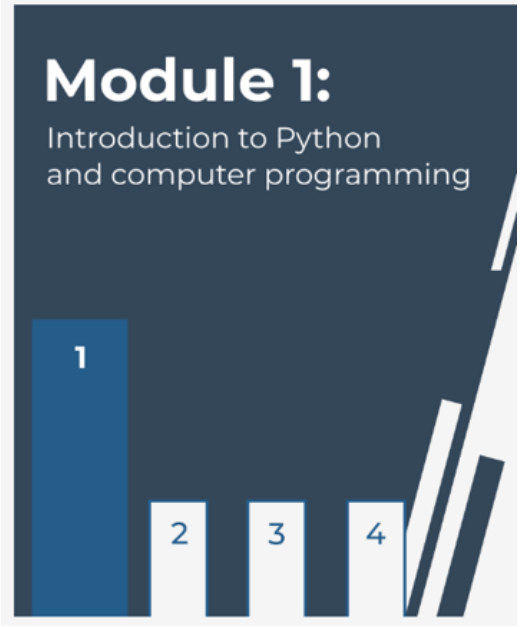


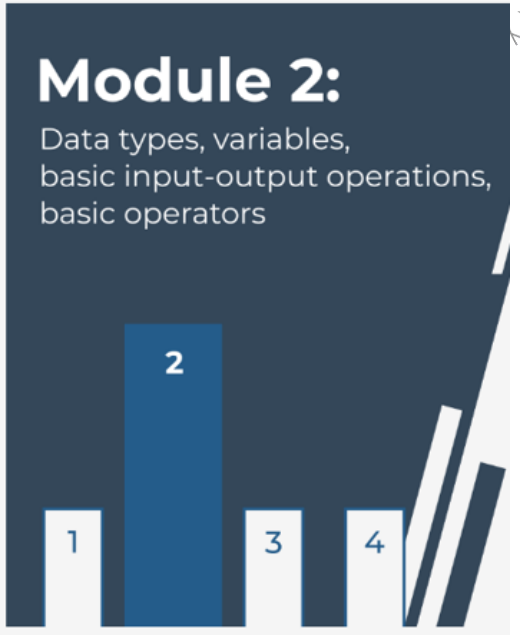


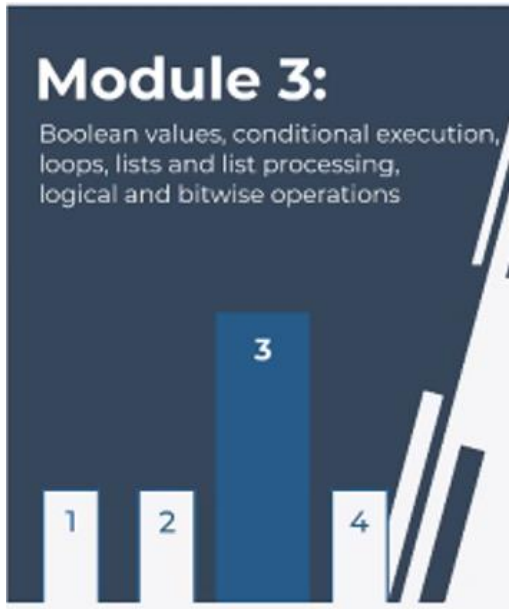


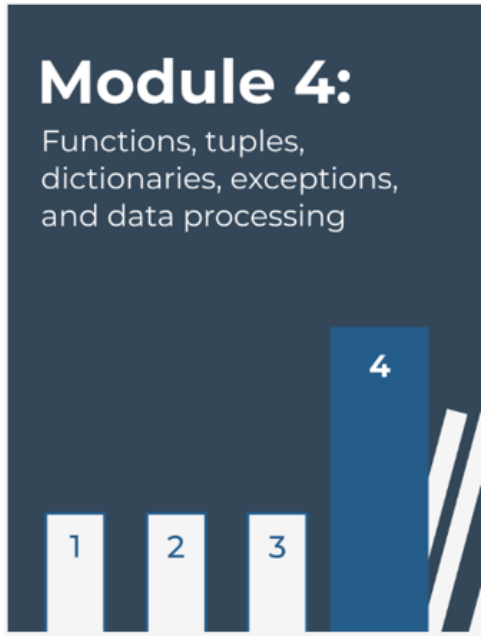
Python 101 & PCEP

1. Chapter 1 Introduction to Computers and Programming 1
2. Chapter 2 Input, Processing, and Output 31
3. Chapter 3 Decision Structures and Boolean Logic 109
4. Chapter 4 Repetition Structures 159
5. Chapter 5 Functions 209
6. Chapter 6 Files and Exceptions 287
7. Chapter 7 Lists and Tuples 343
8. Chapter 8 More About Strings 407
9. Chapter 9 Dictionaries and Sets 439



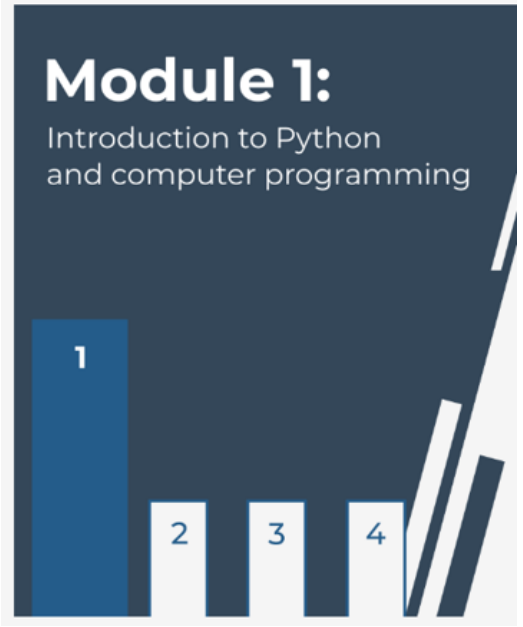


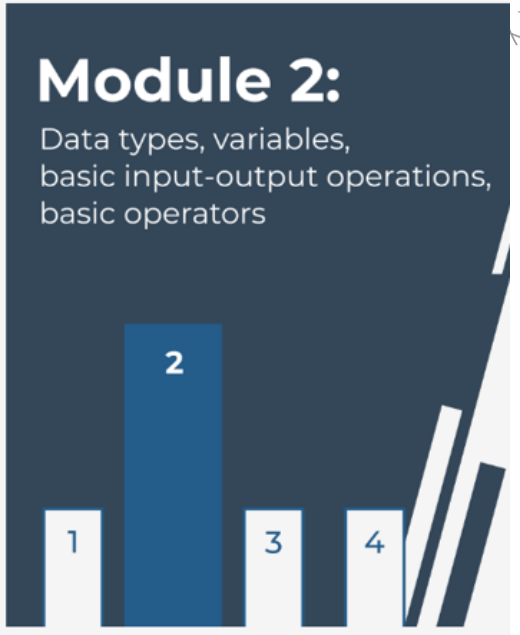


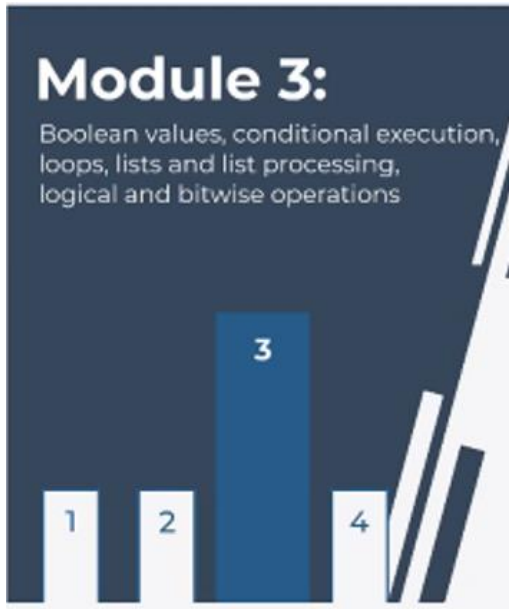


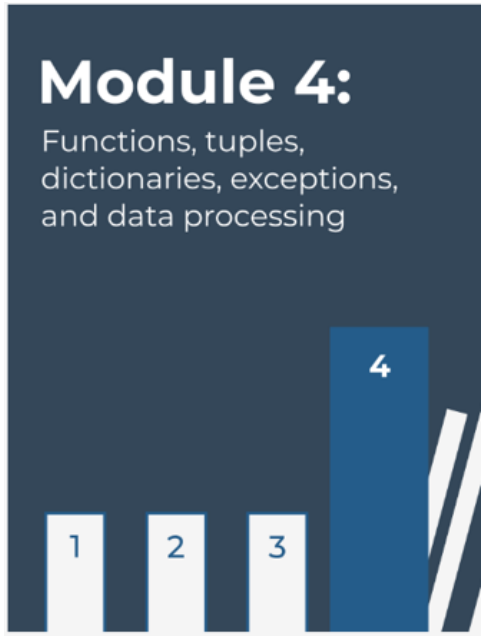
Python 101 to PCEP Mapping

Ch #	Chapter	Module
Ch1.	Introduction to Computers and Programming	Module 1
Ch2.	Input, Processing, and Output	Module 2
Ch3.	Decision Structures and Boolean Logic	Module 3
Ch4.	Repetition Structures	Module 3
Ch5.	Functions	Module 4
Ch6.	Files and Exceptions	Module 4
Ch7.	Lists ad Tuples	Module 3, Module 4
Ch8.	More About Strings	Module 3, Module 4
Ch9.	Dictionaries and Sets	Module 4









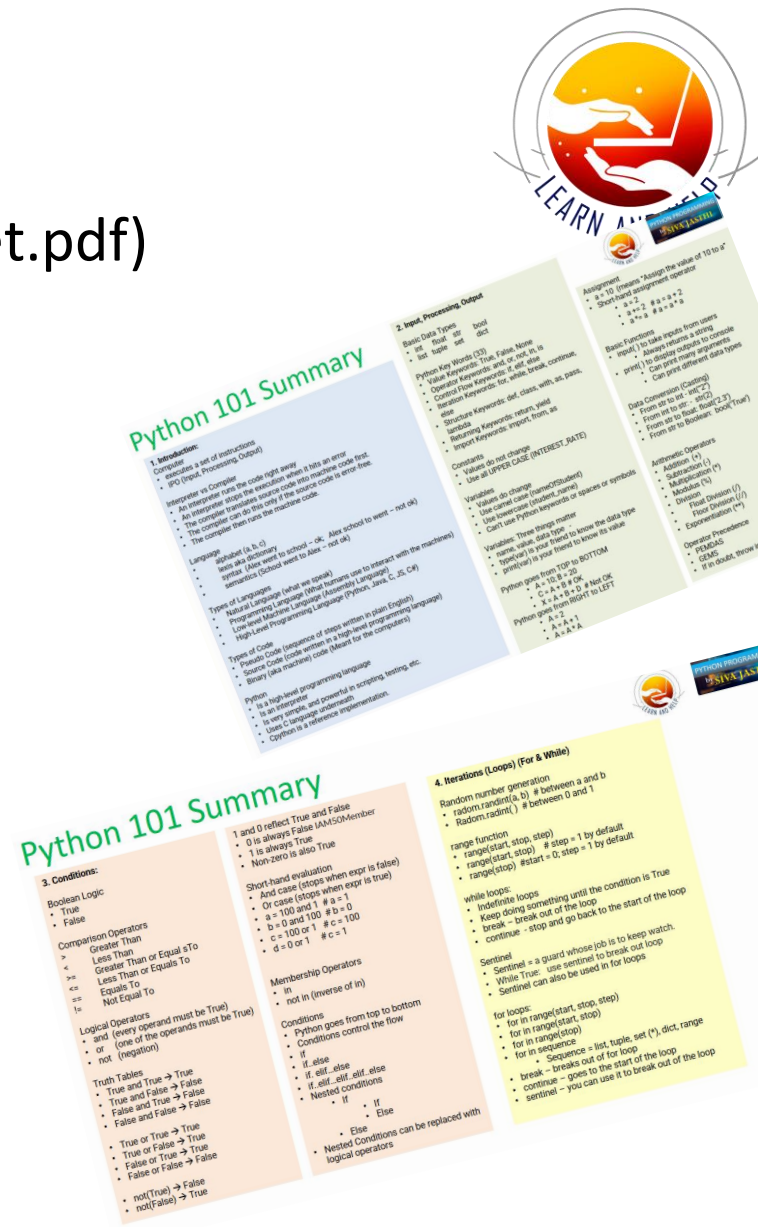
Summary of the course

Cheat Sheets on Chapters (python101_topic_summary_cheatsheet.pdf)

- Review the topics for every chapter
- If any topic is not clear, talk to me.

Some topics worth reading once again

- Interpreter vs compiler
- Binary math and Bitwise Operators
- Else clause in while and for loops
- Exception (try..except..else..finally)
- Difference between functions and methods
- Functions, Scope, global and local variables



Suggested Plan

[1] Focus one module per day (4 * 1 = 4 days = 3 hours each)

- Review the material (Python 101)
- Review the material (PCEP site)
- Take the practice quiz
- Take the Practice Test

Keep taking these quizzes and tests till you score 100% in each of these.

[2] Take the final Summary test (1 day) ~= 1 hour)

- If you score above 90%, you are ready for the PCEP.

[3] Buy this package

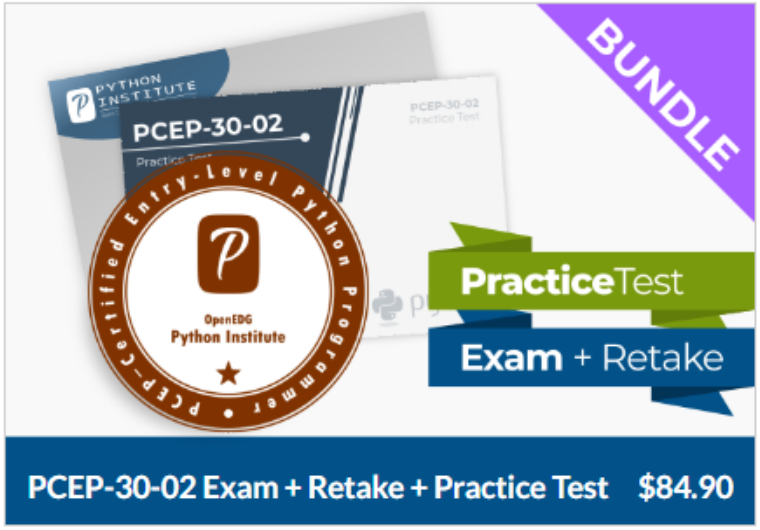
Exam + Retake + Practice Test ($84)

[4] Take the Practice Exam

You can take the practice exam any number of times.

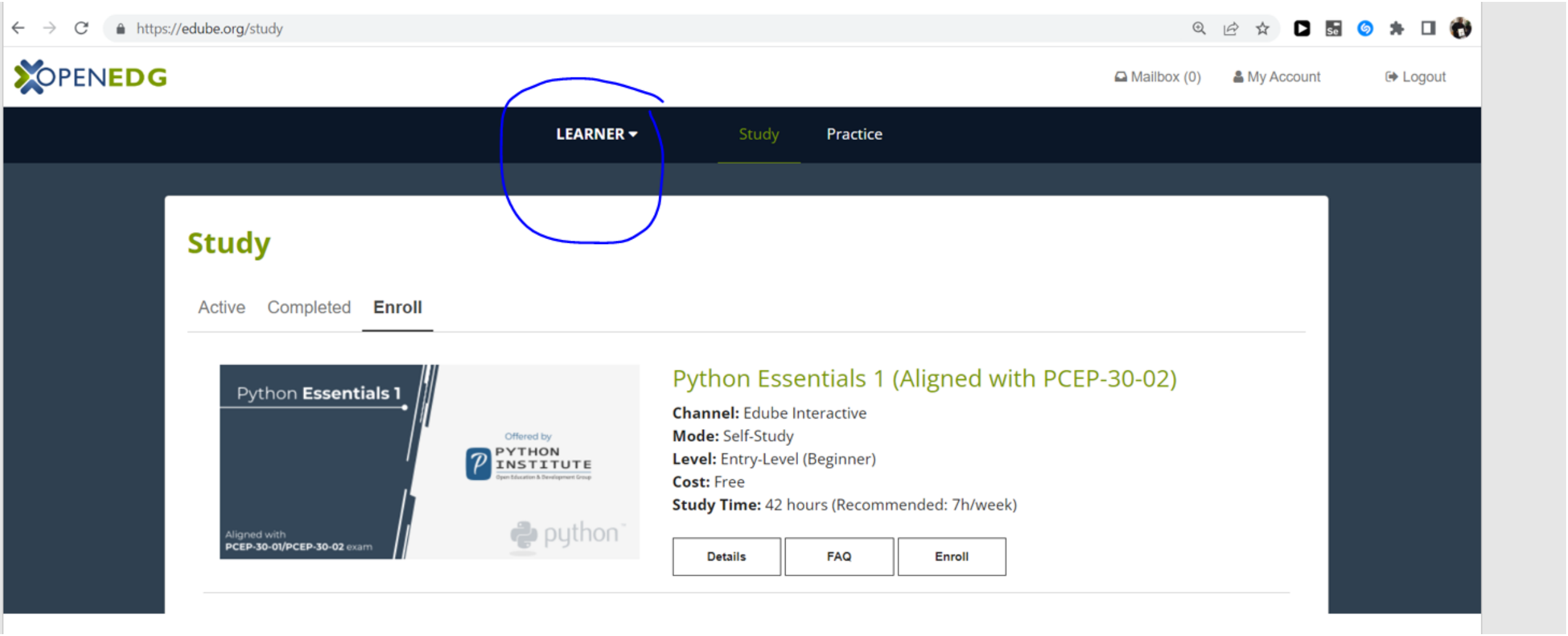
Focus on the concepts and be familiar with the topics.

[5] PCEP exam on the same day

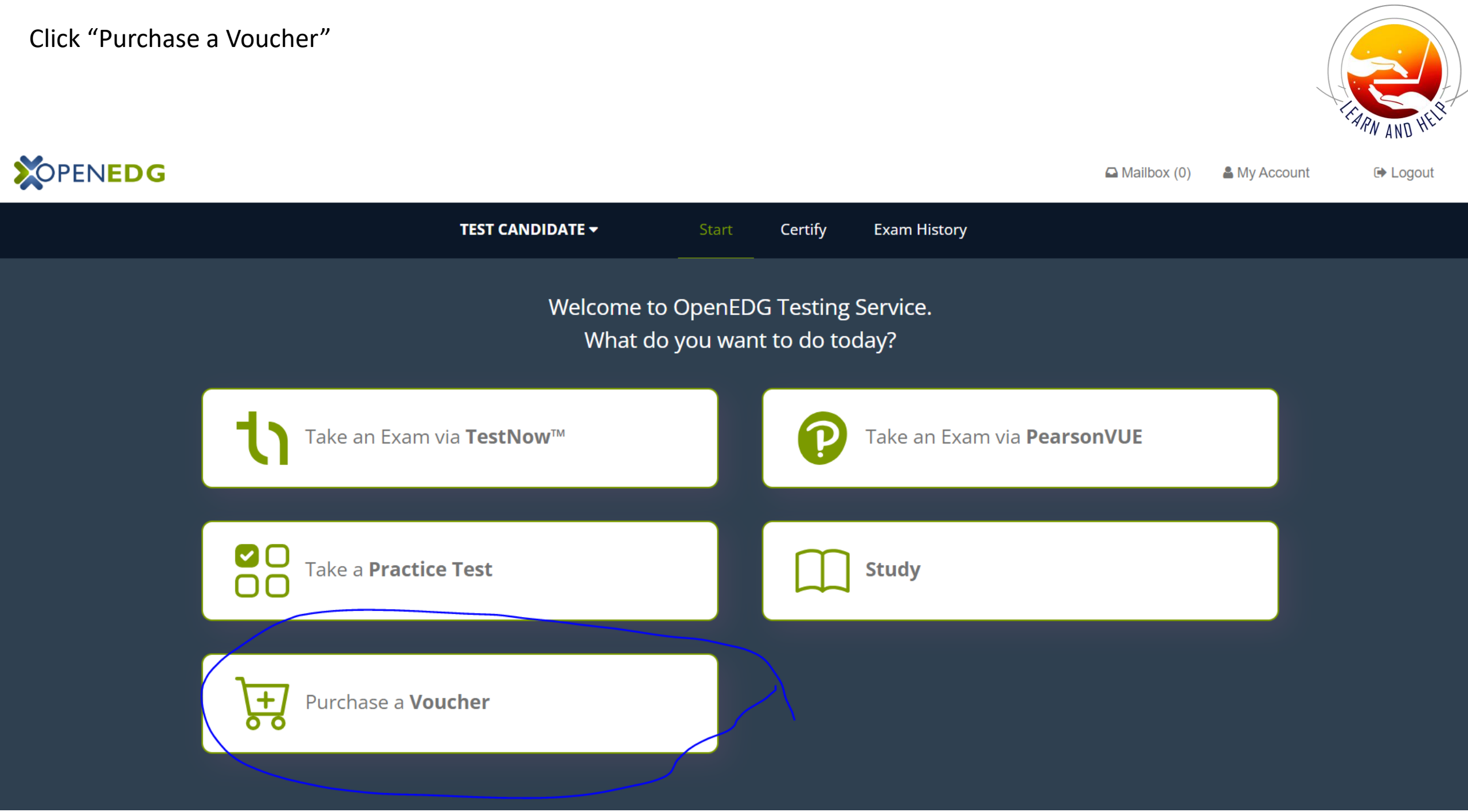


How to buy Exam voucher?

Login to edube.org/study
Change from “Learner → Test Candidate”

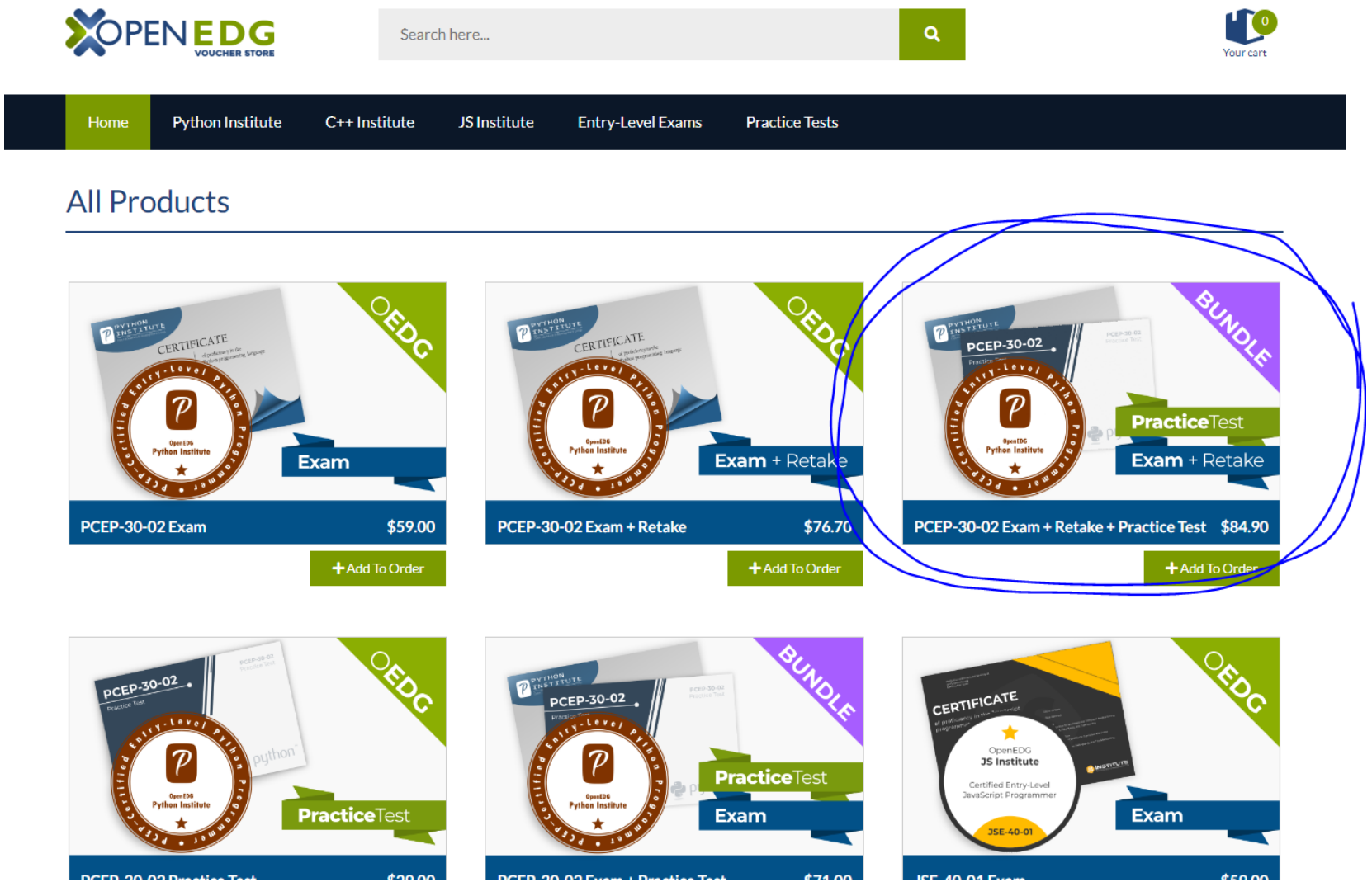


Click “Purchase a Voucher”

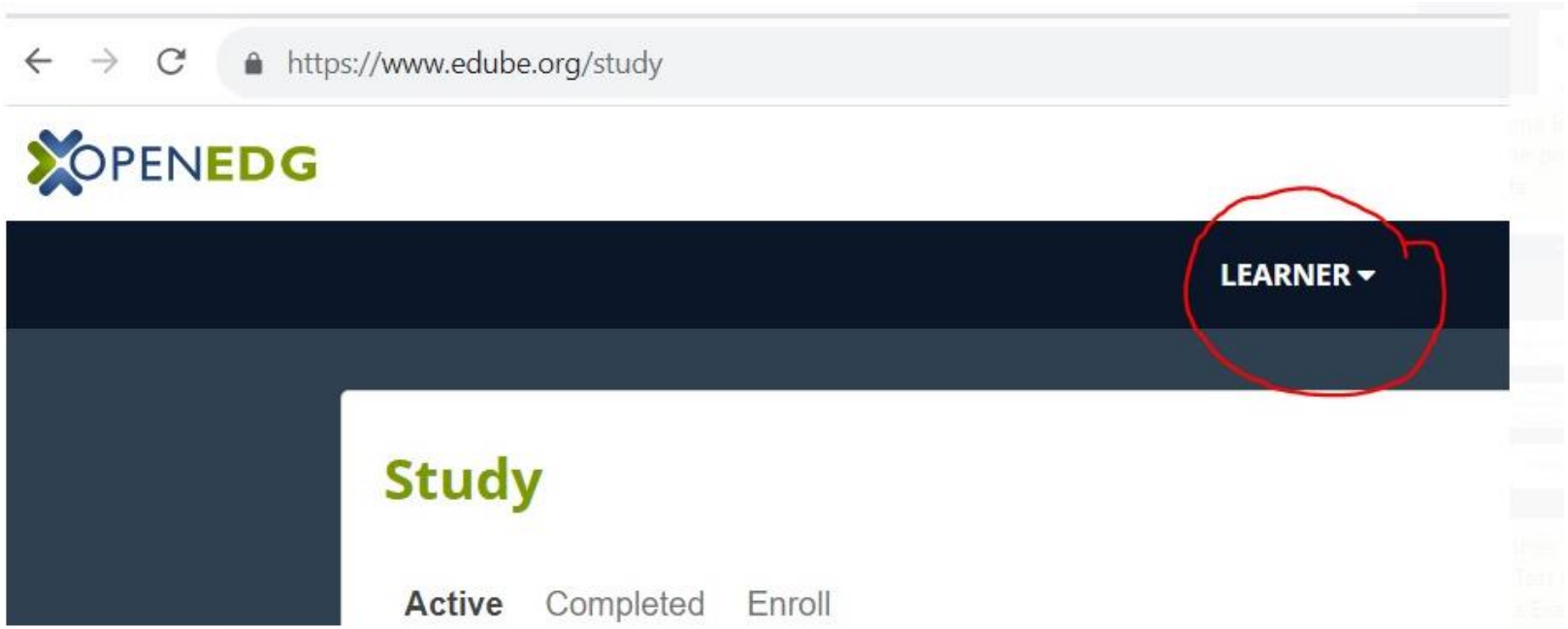


You will come to https://ums.edube.org/store

Buy the voucher for the Bundle (practice test + exam + retake) for $84.90



How to take the test?


https://www.edube.org/study
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Active
Completed
Enroll

OPENEDG
TEST CANDIDATE

TEST CANDIDATE ▾ Start Certify Exam History

Hello, Siva!

What do you want to do today?

Take an Exam via **TestNow™** 2

Take an Exam via **PearsonVUE**

Take a **Practice Test** 1

Study

Purchase a **Voucher**

TestNow™ Pearson VUE Purchase Exam Vouchers

TestNow™

TestNow™ is the OpenEDG online test delivery system that makes it as easy as possible for you to schedule and take your OpenEDG and partner exams.

Take Exam

If you have an exam voucher and wish to begin your exam session immediately, click Take Exam, enter your voucher code and launch the exam.

Take Exam

Assign Vouchers

If you already have an exam voucher, you can assign it to yourself before taking the exam. Your assigned exam vouchers are displayed below and may be used to launch exams via TestNow.

Assign Voucher

Practice

1

Code

Redeem Voucher

2

Voucher Code	Practice Test	Expiration Date	Launches Remaining	Status	
8C686087628F22BCD3E44650F1	PCEP-30-02 Practice Test Compendium	2023-11-27	47	Active	Launch

How to get the certificate?

You would get an email with the subject: " OpenEDG | Congratulations on passing the PCEP-30-02 exam"

OPENEDG

Dear Siva Jasthi,

Congratulations on passing the OpenEDG Python Institute PCEP-30-02 exam, earning your credential, and becoming a **PCEP – Certified Entry-Level Python Programmer**!

Your certification confirms your knowledge in the field of programming and gives you a competitive edge over others in your career. It can help you stand out from other candidates and get your foot in the door.

PCEP – Certified Entry-Level Python Programmer certification is an interim step to the *PCAP – Certified Associate in Python Programming* certification, and the starting point to begin a career in software development, Python programming, and related technologies.

We hope that your achievement will provide you with a solid foundation for you to advance your programming skills and inspire you to attain additional levels of certification.

Please find your score report attached to this email. Your digital certification can be accessed at https://verify.openedg.org/?id=onaD.JPMO.UW40.

You can share your certification with others through your digital transcript by either providing a direct link to your credential, or your unique certification code which employers, colleagues, educators, and classmates can use to verify the status of your credential.

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– VERIFY CERTIFICATE –

PCEP – Certified Entry-Level Python Programmer certification is a professional credential that measures the ability to accomplish coding tasks related to the essentials of programming in the Python language.

A test candidate should demonstrate sufficient knowledge of the universal concepts of computer programming, the syntax and semantics of the Python language as well as the skills in resolving typical implementation challenges with the help of the Python Standard Library.

PCEP – Certified Entry-Level Python Programmer certification shows that the individual is familiar with universal computer programming concepts like data types, containers, functions, conditions, loops, exceptions, as well as Python programming language syntax, semantics, and the runtime environment.

Go to Certification Page for more information.

This certificate is valid for:
Holder's Name: Siva R Jasthi
Credential: PCEP – Certified Entry-Level Python Programmer
Exam Version: PCEP-30-02
Status: Active

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Questions?

Please contact me

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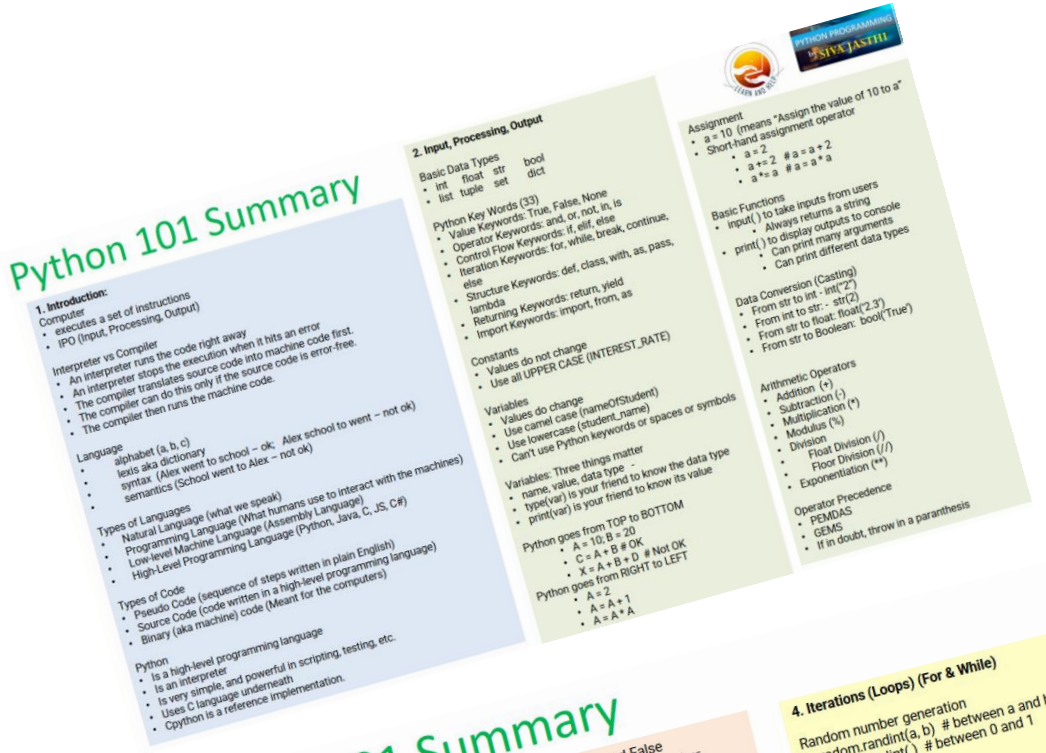


Python 101 Summary

1. Introduction:

Computer
- executes a set of instructions
- IPO (Input, Processing, Output)

Interpreter vs Compiler
- An interpreter runs the code right away
- An interpreter stops the execution when it hits an error
- The compiler translates source code into machine code first.
- The compiler can do this only if the source code is error-free.
- The compiler then runs the machine code.

Language
- alphabet (a, b, c)
- lexis aka dictionary
- syntax (Alex went to school – ok; Alex school to went – not ok)
- semantics (School went to Alex – not ok)

Types of Languages
- Natural Language (what we speak)
- Programming Language (What humans use to interact with the machines)
- Low-level Machine Language (Assembly Language)
- High-Level Programming Language (Python, Java, C, JS, C#)

Types of Code
- Pseudo Code (sequence of steps written in a high-level programming language)
- Source Code (code written in a high-level programming language)
- Binary (aka machine) code (Meant for the computers)

Python
- Is a high-level programming language
- Is an interpreter
- Is very simple, and powerful in scripting, testing, etc.
- Uses C language underneath
- Cpython is a reference implementation.

2. Input, Processing, Output

Basic Data Types
- int float str bool
- list tuple set dict

Python Key Words (33)
- Value Keywords: True, False, None
- Operator Keywords: and, or, not, in, is
- Control Flow Keywords: if, elif, else
- Iteration Keywords: for, while, break, continue, else
- Structure Keywords: def, class, with, as, pass, lambda
- Returning Keywords: return, yield
- Import Keywords: import, from, as

Constants
- Values do not change
- Use all UPPER CASE (INTEREST_RATE)

Variables
- Values do change
- Use camel case (nameOfStudent)
- Use lowercase (student_name)
- Can't use Python keywords or spaces or symbols

Variables: Three things matter
- name, value, data type
- type(var) is your friend to know the data type
- print(var) is your friend to know its value

Python goes from TOP to BOTTOM
- A = 10; B = 20
- C = A + B # OK
- X = A + B + D # Not OK

Python goes from RIGHT to LEFT
- A = 2
- A = A + 1
- A = A * A

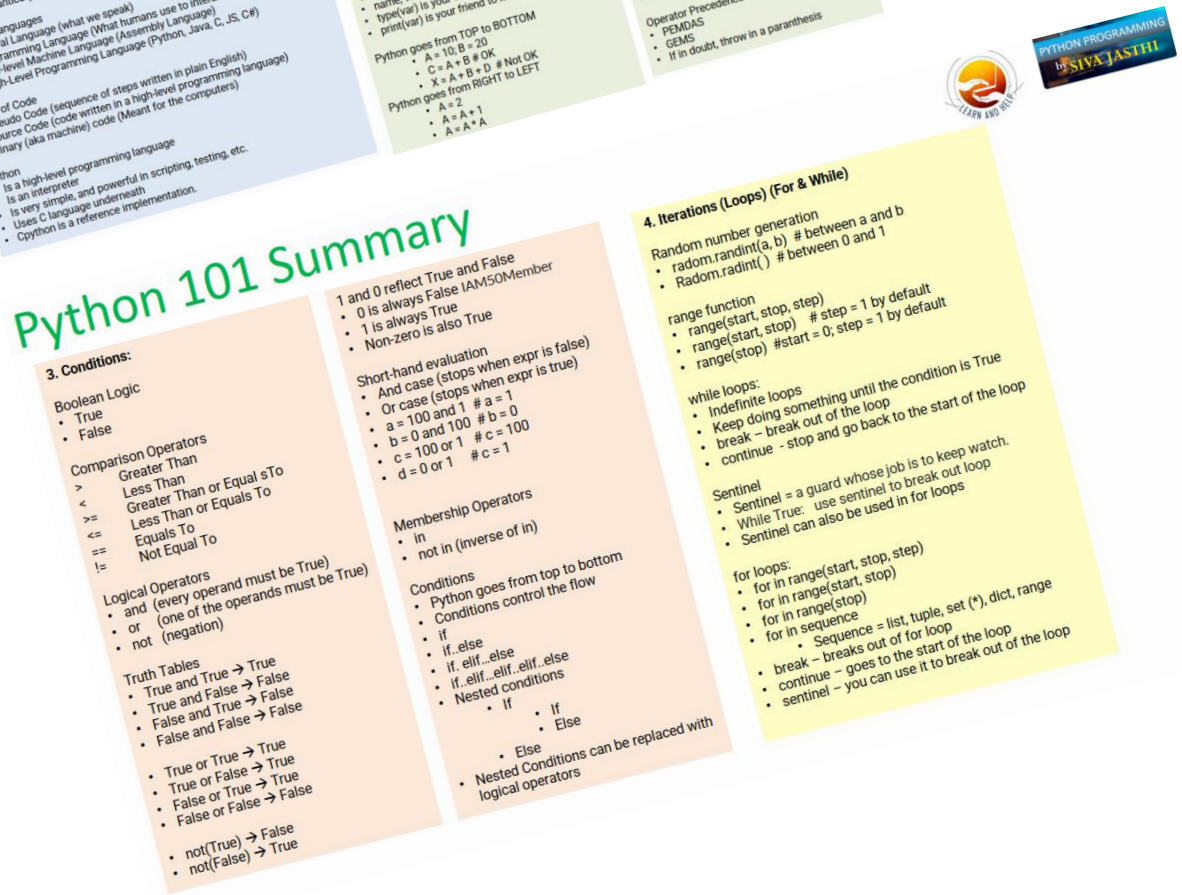
Assignment
- a = 10 (means "Assign the value of 10 to a"
- Short-hand assignment operator
  - a = 2
  - a += 2 # a = a + 2
  - a *= a # a = a * a

Basic Functions
- input( ) to take inputs from users
  - Always returns a string
- print( ) to display outputs to console
  - Can print many arguments
  - Can print different data types

Data Conversion (Casting)
- From str to int - int("2")
- From int to str: - str(2)
- From str to float: float('2.3')
- From str to Boolean: bool('True')

Arithmetic Operators
- Addition (+)
- Subtraction (-)
- Multiplication (*)
- Modulus (%)
- Division
  - Float Division (/)
  - Floor Division (//)
- Exponentiation (**)

Operator Precedence
- PEMDAS
- GEMS
- If in doubt, throw in a paranthesis



Python 101 Summary

3. Conditions:

Boolean Logic
- True
- False

Comparison Operators
- > Greater Than
- < Less Than
- >= Greater Than or Equal sTo
- <= Less Than or Equals To
- == Equals To
- != Not Equal To

Logical Operators
- and (every operand must be True)
- or (one of the operands must be True)
- not (negation)

Truth Tables
- True and True → True
- True and False → False
- False and True → False
- False and False → False

- True or True → True
- True or False → True
- False or True → True
- False or False → False

- not(True) → False
- not(False) → True

1 and 0 reflect True and False
- 0 is always False IAM50Member
- 1 is always True
- Non-zero is also True

Short-hand evaluation
- And case (stops when expr is false)
- Or case (stops when expr is true)
- a = 100 and 1 # a = 1
- b = 0 and 100 # b = 0
- c = 100 or 1 # c = 100
- d = 0 or 1 # c = 1

Membership Operators
- in
- not in (inverse of in)

Conditions
- Python goes from top to bottom
- Conditions control the flow
- if
- if..else
- if..elif...else
- if..elif...elif..elif..else
- Nested conditions
  - If
    - If
    - Else
  - Else
- Nested Conditions can be replaced with logical operators

4. Iterations (Loops) (For & While)

Random number generation
- radom.randint(a, b) # between a and b
- Radom.radint( ) # between 0 and 1

range function
- range(start, stop, step)
- range(start, stop) # step = 1 by default
- range(stop) #start = 0; step = 1 by default

while loops:
- Indefinite loops
- Keep doing something until the condition is True
- break – break out of the loop
- continue - stop and go back to the start of the loop

Sentinel
- Sentinel = a guard whose job is to keep watch.
- While True: use sentinel to break out loop
- Sentinel can also be used in for loops

for loops:
- for in range(start, stop, step)
- for in range(start, stop)
- for in range(stop)
- for in sequence
  - Sequence = list, tuple, set (*), dict, range
- break – breaks out of for loop
- continue – goes to the start of the loop
- sentinel – you can use it to break out of the loop

Thank You.